

Chapter 17 Annelids and Allied Taxa

Introduction

- The evolutionary advent of **metamerism** (segmentation) allowed for greater complexity in structure and function.
- Metamerism increased burrowing efficiency by permitting independent movement of separate segments.
- It allowed for the evolution of a more sophisticated nervous system and provided a safety factor through redundancy of body parts.
- The phyla Annelida, Arthropoda, and Chordata likely represent three separate evolutionary origins of metamerism.
- Annelids, Echiurans, and Sipunculans are coelomate protostomes belonging to subgroup **Lophotrochozoa**.
- They share spiral mosaic cleavage, mesoderm formation from the 4d cell, schizocoelous coelom formation, and a **trochophore** larva.



Figure 1: Cladogram of annelids, showing the appearance of shared derived characters that specify the five monophyletic groups. The Acanthobdellida and the Branchiobdellida are two small groups that are placed with the Hirudinea (“true” leeches), within a single taxon, the Hirudinida. This clade has several synapomorphies: tendency toward reduction of septal walls, the appearance of a posterior sucker, and the subdivision of body segments by superficial annuli. Note also that, according to this scheme, the Oligochaeta have no defining synapomorphies; that is, they are defined solely by retention of plesiomorphies (retained primitive characters), and thus might be paraphyletic.

- **Figure 17.1 Explanation:** This cladogram illustrates the relationships within Annelida. It shows the appearance of shared derived characters for the five monophyletic groups. Note that **Hirudinida** (leeches) includes “true” leeches and related groups, characterized by reduced septal walls and posterior suckers. **Oligochaeta** is depicted as potentially paraphyletic, defined by retained primitive characters.

Phylum Annelida

- Phylum **Annelida** consists of segmented worms, including marine polychaetes, earthworms, freshwater oligochaetes, and leeches.
- Annelids are defined by **metamerism**: the division of the body into a series of similar segments (metameres) arranged linearly and marked externally by **annuli**.
- Most annelids bear tiny chitinous bristles called **setae** (except leeches).
 - Short needlelike setae anchor segments during locomotion.
 - Long hairlike setae aid in swimming.

Body Plan

- The annelid head typically consists of a two-part head: a **prostomium** and a **peristomium**.
- The segmented body is followed by a terminal portion called the **pygidium** bearing an anus.
- The head and pygidium are not considered true segments.
- New segments differentiate just in front of the pygidium; thus, the oldest segments are anterior and the youngest are posterior.
- The **coelom** develops as a split in the mesoderm (**schizocoel**), forming paired compartments in each segment.
- **Peritoneum** (mesodermal epithelium) lines the body wall and forms dorsal and ventral **mesenteries**.
- **Septa** separate segments, perforated by the gut and longitudinal blood vessels.
- The coelom is filled with fluid and serves as a **hydrostatic skeleton** (except in leeches).
 - Contraction of longitudinal muscles shortens and widens the segment.
 - Contraction of circular muscles lengthens and thins the segment.
 - Peristaltic contractions allow for crawling and burrowing.



Figure 2: **Nereis virens** (A–D) and **Nereis diversicolor** (E) are errant polychaetes. A, Anterior end, with pharynx everted. B, External structure. C, Posterior end. D, Generalized transverse section through region of the intestine. E, In this photo of a live **N. diversicolor**, note the well-defined segments, the lobed parapodia, and the prostomium with tentacles.

- **Figure 17.2 Explanation:** *Nereis* (clam worm) exemplifies the errant polychaete body plan. It possesses a distinct head with a **prostomium** and **peristomium**, and paired appendages called **parapodia** on most segments. The parapodia bear setae and are used for locomotion and respiration.



Figure 3: Annelid body plan.

- **Figure 17.3 Explanation:** This diagram shows the generalized annelid body plan in cross-section. It highlights the **coelom** divided by **septa**, the arrangement of circular and longitudinal muscles, the dorsal and ventral blood vessels, and the nerve cord. The **peritoneum** lines the coelom and covers the internal organs.

Class Polychaeta

- **Polychaeta** is the largest class, mostly marine.
- They differ from other annelids by having a well-differentiated head with specialized sense organs, paired **parapodia** on most segments, and no **clitellum**.
- Polychaetes are often divided into:
 - **Sedentary polychaetes:** Spend time in tubes or permanent burrows; often have elaborate devices for feeding and respiration.
 - **Errant polychaetes:** Free-moving, active burrowers, crawlers, or pelagic forms; often predatory with jaws.



Figure 4: Tube-dwelling sedentary polychaetes. A, Christmas-tree worm, **Spirobranchus giganteus**, live in a calcareous tube. On its head are two whorls of modified tentacles (radioles) used to collect suspended food particles from the surrounding water. Notice the finely branched filters visible on the edge of one radiole. B, Sabellid polychaetes, **Bispira brunnea**, live in leathery tubes.

- **Figure 17.4 Explanation:** Examples of tube-dwelling sedentary polychaetes. The Christmas-tree worm (**Spirobranchus**) and Sabellid polychaetes (**Bispira**) use modified tentacles called **radioles** for filter feeding and respiration.

Form and Function

- **Prostomium:** Often bears eyes, tentacles, and sensory palps.
- **Peristomium:** Surrounds the mouth; may bear setae, palps, or chitinous jaws.
- **Parapodia:** Paired appendages on segments, used for crawling, swimming, or anchoring. They usually serve as the chief respiratory organs.
- Some polychaetes have gills (e.g., **Amphitrite**, **Arenicola**).



Figure 5: **Amphitrite**, which builds its tubes in mud or sand, extends long grooved tentacles out over the mud to pick up bits of organic matter. The smallest particles are moved along food grooves by cilia, larger particles by peristaltic movement. Its plumelike gills are blood red. A, Section through exploratory end of tentacle. B, Section through tentacle in area adhering to substratum. C, Section showing ciliary groove. D, Particle being carried toward mouth.

- **Figure 17.5 Explanation: Amphitrite** is a deposit feeder. It extends long, grooved tentacles over the substrate to collect organic particles. Cilia move small particles along food grooves to the mouth, while larger particles are moved by peristalsis.

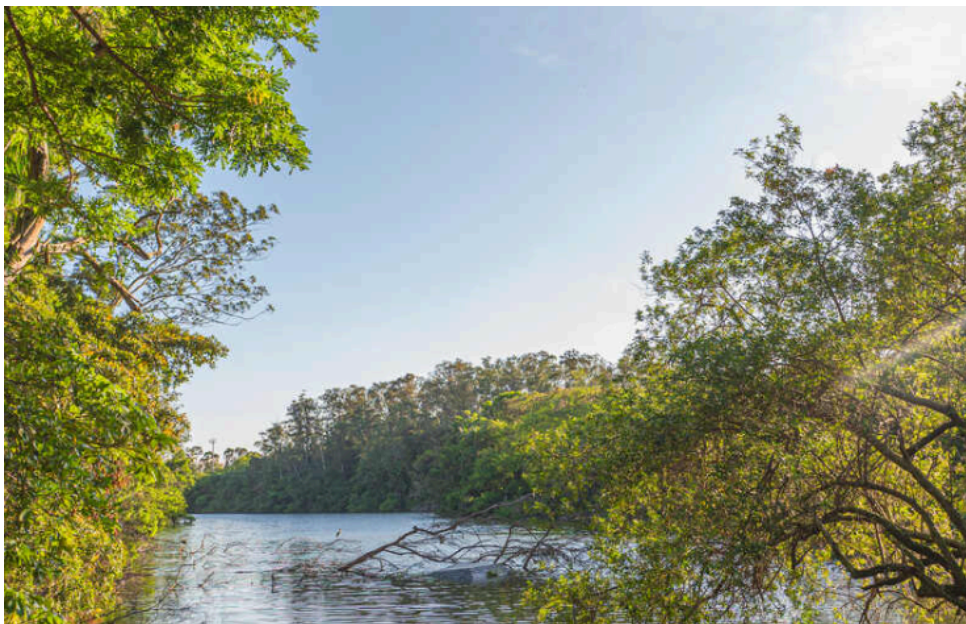


Figure 6: **Arenicola**, the lugworm, lives in a U-shaped burrow in intertidal mudflats. It burrows by successive eversions and retractions of its proboscis. By peristaltic movements it keeps water filtering through the sand. The worm then ingests the food-laden sand.

- **Figure 17.6 Explanation: Arenicola** (lugworm) lives in a U-shaped burrow. It maintains water flow by peristalsis and ingests food-laden sand at the front of the burrow, combining suspension and deposit feeding.

Nutrition

- Digestive system: Foregut (stomodeum, pharynx, anterior esophagus), midgut (secretion and absorption), and hindgut (connects to anus).
- Errant forms are often predators/scavengers with jaws.
- Sedentary forms are suspension feeders (ciliary feeders) or deposit feeders.

Circulation and Respiration

- Respiration: Via parapodia, gills, or body surface.
- Circulation: Closed system in most. Dorsal vessel carries blood anteriorly; ventral vessel carries blood posteriorly.
- Respiratory pigments: Hemoglobin, chlorocruorin, or hemerythrin.

Excretion

- Excretory organs: Protonephridia or **metanephridia**.
- Typically one pair of metanephridia per segment.
- The inner end (**nephrostome**) opens into the coelom; the duct leads to the exterior.

Nervous System and Sense Organs

- Dorsal cerebral ganglia connect to a subpharyngeal ganglion via a circumpharyngeal connective.
- Double ventral nerve cord with metameric ganglia.
- Sense organs: Eyes (from eyespots to image-resolving eyes in Alciopidae), **nuchal organs** (chemoreceptive pits), statocysts.

Reproduction and Development

- Usually separate sexes; no permanent sex organs.
- Gonads are temporary swellings of the peritoneum; gametes shed into coelom.
- External fertilization; **trochophore** larva.
- **Epitoky**: Formation of a reproductive individual (**epitoke**) which differs from the non-reproductive form (**atoke**).
 - Swarming ensures synchronous maturation and fertilization.



Figure 7: **Eunice viridis**, the Samoan palolo worm. The posterior segments make up the epitokal region, consisting of segments packed with gametes. Each segment has an eyespot on the ventral side. Once a year the worms swarm, and the epitokes detach, rise to the surface, and discharge their ripe gametes, leaving the water milky. By the next breeding season, the epitokes are regenerated.

- **Figure 17.7 Explanation: Eunice viridis** (palolo worm) exhibits epitoky. The posterior segments become the epitoke, packed with gametes, which breaks off and swarms to the surface to release gametes.

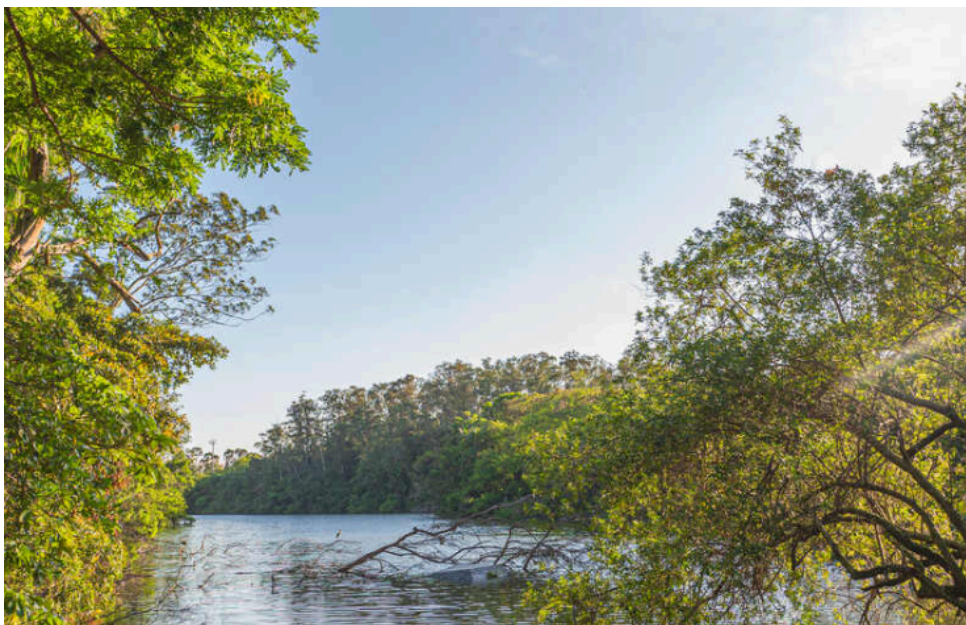


Figure 8: Rather than transforming a portion of its body into an epitoke, **Autolytus prolifer** asexually buds off complete worms from its posterior end that become sexual epitokes.

- **Figure 17.8 Explanation: Autolytus prolifer** reproduces by asexual budding, where complete worms (epitokes) bud off from the posterior end.

Representative Polychaetes

- **Clam Worms (Nereis)**: Errant, predatory, active at night. Distinct head, parapodia with two lobes (**notopodium** and **neuropodium**).
- **Scale Worms**: Flattened bodies covered with broad scales (modified dorsal parapodia). Carnivorous or commensal.



Figure 9: A scale worm, **Hesperonoe adventor**, normally lives as a commensal in the tubes of **Urechis** (phylum Echiura).

- **Figure 17.9 Explanation: Hesperonoe adventor** is a scale worm, often found living commensally with other organisms.
- **Fireworms**: Hollow, brittle setae contain poisonous secretion. Feed on cnidarians.

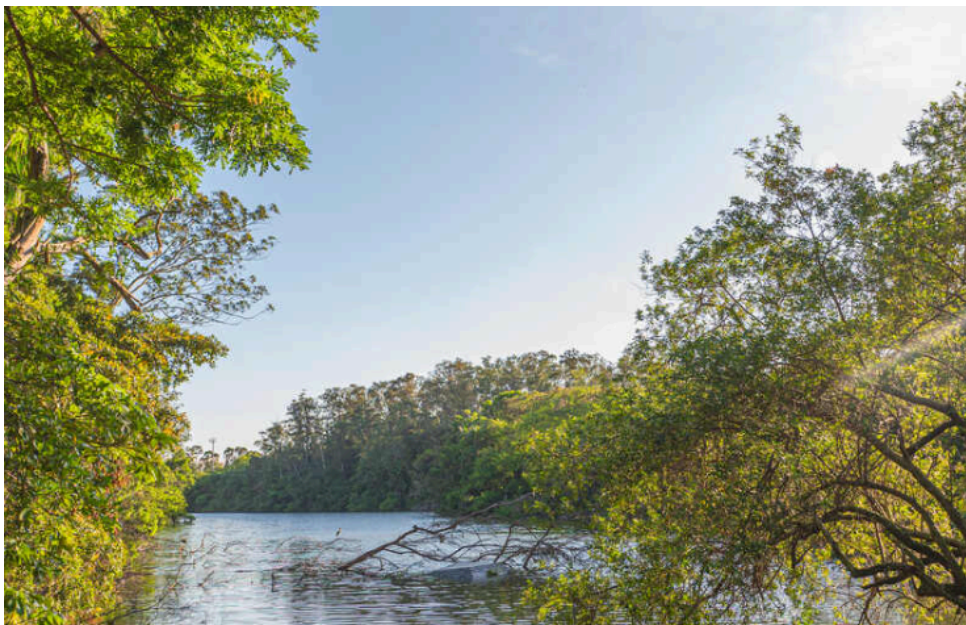


Figure 10: A fireworm, **Hermodice carunculata**, feeds on gorgonians and stony corals. Its setae are like tiny glass fibers and serve to ward off predators.

- **Figure 17.10 Explanation: *Hermodice carunculata*** (fireworm) has setae that can puncture skin and cause irritation.
- **Tubeworms:** Secrete tubes (parchment, calcareous, sand/mucus).
 - **Fanworms:** Use **radioles** (feathery arms) for ciliary feeding.



Figure 11: **Sabella**, a polychaete ciliary feeder, extends its crown of feeding radioles from its leathery secreted tube, reinforced with sand and debris. A, Anterior view of the crown. Cilia direct small food particles along grooved radioles to mouth and discard larger particles. Sand grains are directed to storage sacs and later are used in tube building. B, Distal portion of radiole showing ciliary tracts of pinnules and food grooves.

- **Figure 17.11 Explanation: *Sabella*** (fanworm) uses its radioles to trap food particles in mucus and transport them to the mouth via ciliated grooves. It sorts particles by size.
 - **Parchment Worm (*Chaetopterus*):** Lives in U-shaped tube; uses mucous net to filter food.



Figure 12: **Chaetopterus**, a sedentary polychaete, lives in a U-shaped tube in the sea bottom. It pumps water through the parchmentlike tube (of which one-half has been cut away here) with its three pistonlike fans. The fans beat 60 times per minute to keep water currents moving. The winglike notopodia of the twelfth segment continuously secrete a mucous net that strains out food particles. As the net fills with food, the food cup rolls it into a ball, and when the ball is large enough (about 3 mm), the food cup bends forward and deposits the ball in a ciliated groove to be carried to the mouth and swallowed.

- **Figure 17.12 Explanation: Chaetopterus** pumps water through its tube using modified parapodia ("fans"). It secretes a mucous net from the 12th segment notopodia to trap food, which is then rolled into a ball and swallowed.

Clade Siboglinidae (Pogonophorans)

- Formerly Phylum Pogonophora (beardworms). Now placed in Class Polychaeta.
- Deep-sea, sessile, tube-dwelling.
- Body: Short anterior **forepart**, long slender **trunk**, small segmented **opisthosoma**.
- **Tentacles** ("beard") on cephalic lobe; hollow extensions of coelom with pinnules.



Figure 13: Diagram of a typical siboglinid. A, External features. In life, the body is much more elongated than shown in this diagram. B, Position in tube.

- **Figure 17.13 Explanation:** Diagram of a siboglinid showing the forepart with tentacles, the long trunk, and the segmented opisthosoma which anchors the worm in its tube.



Figure 14: Cross section of tentacular crown of siboglinid **Lamellisabella**. Tentacles arise from ventral side of forepart at base of cephalic lobe. Tentacles (which vary in number in different species) enclose a cylindrical space, with the pinnules forming a kind of nutrient uptake network. Food molecules may be absorbed into the blood supply of tentacles and pinnules.

- **Figure 17.14 Explanation:** Cross-section of the tentacular crown. The tentacles enclose a space where pinnules project, increasing surface area for nutrient uptake.
- **Nutrition:** No mouth or digestive tract.
 - Absorb dissolved nutrients.

- Mutualistic association with **chemoautotrophic bacteria** in an organ called the **trophosome**. Bacteria oxidize hydrogen sulfide to produce organic compounds.
- **Vestimentiferans** (e.g., **Riftia pachyptila**): Giant beardworms found at hydrothermal vents.



Figure 15: A colony of giant beardworms (vestimentiferans, clade Siboglinidae) at great depth near a hydrothermal vent along the Galápagos Trench, eastern Pacific Ocean.

- **Figure 17.15 Explanation: Riftia pachyptila** (giant beardworms) thrive near hydrothermal vents, relying on their symbiotic bacteria to utilize the abundant hydrogen sulfide.

Clade Clitellata

- Contains Class **Oligochaeta** (earthworms) and Class **Hirudinida** (leeches).
- Characterized by a **clitellum**: a ring of secretory cells used in reproduction.
- Lack parapodia.
- Hermaphroditic (monoecious) with direct development (no trochophore larva).

Class Oligochaeta

- Includes earthworms and freshwater worms.
- Bear **setae**, but fewer than polychaetes; no parapodia.

Form and Function (**Lumbricus terrestris**)

- Scavengers, burrowing in moist soil.
- Locomotion via peristalsis: Alternating contraction of circular and longitudinal muscles, using setae to anchor.



Figure 16: Earthworm anatomy. A, External features, lateral view. B, Internal structure of anterior portion of worm. C, Generalized transverse section through region posterior to clitellum. D, Portion of epidermis showing sensory, glandular, and epithelial cells.

- **Figure 17.16 Explanation:** Anatomy of an earthworm. Note the **clitellum**, the lack of parapodia, and the internal arrangement of organs including the **typhlosole** in the intestine which increases surface area.

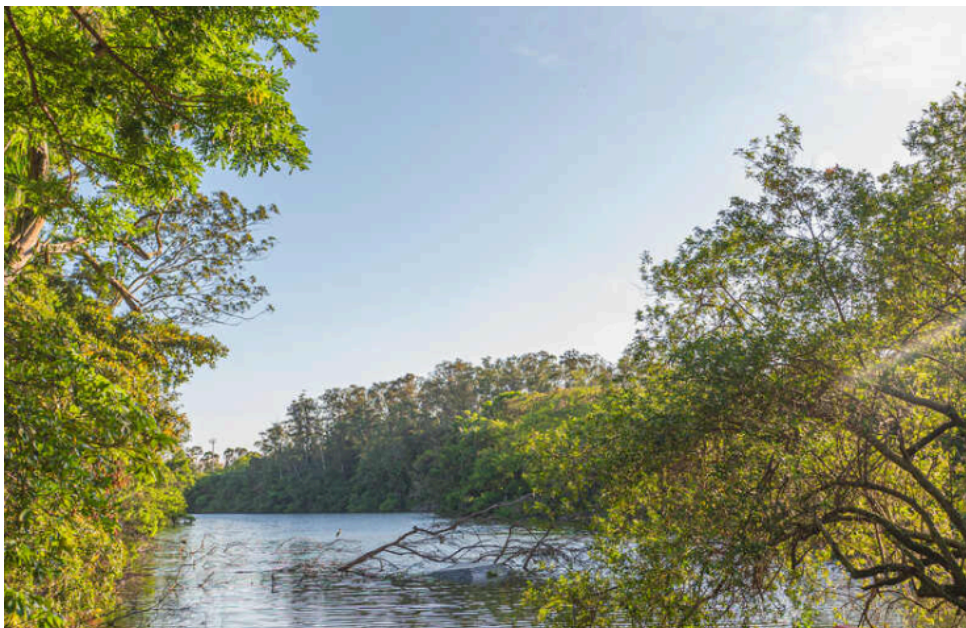


Figure 17: Seta with its muscle attachments showing relation to adjacent structures. Setae lost by wear are replaced by new ones, which develop from formative cells.

- **Figure 17.17 Explanation:** Detailed view of a seta. Muscles allow the seta to be protracted (to anchor) or retracted.

Nutrition

- Scavengers feeding on decaying organic matter.

- Digestive tract: Mouth -> Pharynx (sucking) -> Esophagus (**calciferous glands** regulate calcium/pH) -> **Crop** (storage) -> **Gizzard** (grinding) -> **Intestine** (digestion/absorption).
- **Typhlosole**: Dorsal infolding of intestine to increase surface area.
- **Chloragogen tissue**: Surrounds intestine; synthesizes glycogen/fat (liver-like function) and aids in excretion.

Circulation and Respiration

- Double transport system: Coelomic fluid and closed circulatory system.
- Five main blood trunks; **Dorsal vessel** (pumping organ) and **Ventral vessel** (aorta).
- **Aortic arches**: Maintain blood pressure.
- Blood contains **hemoglobin**.
- Respiration occurs across the moist skin.

Excretion

- **Metanephridia**: Paired in each segment (except first three and last).
- **Nephrostome** draws in coelomic fluid; tubule reabsorbs salts/nutrients; waste discharged via **nephridiopore**.
- Excrete ammonia (aquatic) or urea (terrestrial).

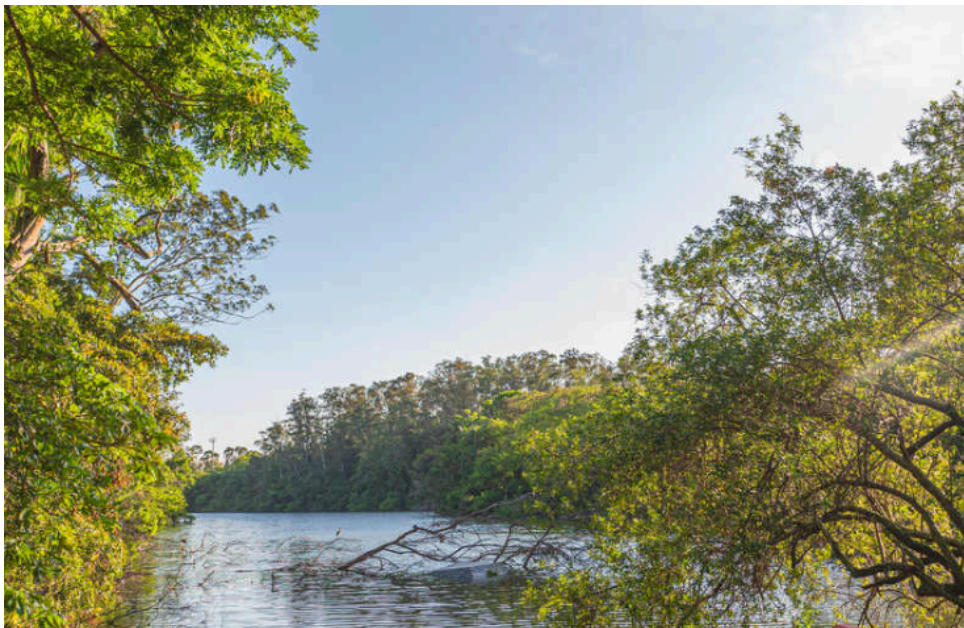


Figure 18: Nephridium of earthworm. Wastes are drawn into the ciliated nephrostome in one segment, then passed through the loops of the nephridium, and expelled through the nephridiopore of the next segment.

- **Figure 17.18 Explanation**: Structure of a metanephridium. It spans two segments: the nephrostome is in the anterior segment, while the tubule and bladder are in the posterior segment.

Nervous System and Sense Organs

- Central nervous system: Pair of cerebral ganglia ("brain"), circumpharyngeal connectives, double ventral nerve cord with fused segmental ganglia.
- **Giant axons** (giant fibers) in nerve cord allow for rapid conduction and simultaneous muscle contraction (escape reflex).



Figure 19: Anterior portion of earthworm and its nervous system. Note concentration of sensory endings in this region.

- **Figure 17.19 Explanation:** Anterior nervous system of an earthworm, showing the cerebral ganglia and the concentration of sensory endings in the prostomium.



Figure 20: Portion of nerve cord of earthworm showing arrangement of simple reflex arc (in foreground) and the three dorsal giant fibers that are adapted for rapid reflexes and escape movements. Ordinary crawling involves a succession of reflex acts, the stretching of one segment stimulating the next to stretch. Impulses are transmitted much faster in giant fibers than in regular nerves so that all segments can contract simultaneously when quick withdrawal into a burrow is necessary.

- **Figure 17.20 Explanation:** Cross-section of the nerve cord showing the dorsal giant fibers. These fibers facilitate rapid escape reflexes.

- Sense organs: No eyes, but lens-shaped photoreceptors in epidermis; chemoreceptors; tactile nerve endings.

Reproduction and Development

- Monoecious (hermaphroditic).
- Reproductive system in segments 9-15 (**Lumbricus**).
- Mating: Ventral surfaces together, held by mucus from **clitellum**. Sperm exchange.
- **Cocoon** formation: Clitellum secretes cocoon; eggs and sperm deposited into it. Fertilization and embryogenesis occur inside.
- Direct development: Young worms hatch from cocoon.



Figure 21: Earthworm copulation and formation of egg cocoons. A, Mutual insemination; sperm from genital pore (segment 15) pass along seminal grooves to seminal receptacles (segments 9 and 10) of each mate. B and C, After worms separate, the clitellum secretes first a mucous tube and then a tough band that forms a cocoon. The developing cocoon passes forward to receive eggs from oviducts and sperm from seminal receptacles. D, As cocoon slips off over anterior end, its ends close and seal. E, Cocoon is deposited near burrow entrance. F, Young worms emerge in 2 to 3 weeks. G, Two earthworms in copulation. Their anterior ends point in opposite directions as their ventral surfaces are held together by mucous bands secreted by the clitella.

- **Figure 17.21 Explanation:** The reproductive process of earthworms. (A) Copulation and sperm exchange. (B-E) Formation of the cocoon by the clitellum and deposition of gametes. (F) Emergence of young worms.

Representative Oligochaetes

- Freshwater forms: Smaller, more conspicuous setae, more mobile.
- Examples: **Aelosoma**, **Stylaria**, **Tubifex** (host for whirling disease parasite).



Figure 22: Some freshwater oligochaetes. A, **Stylaria** has the prostomium drawn out into a long snout. B, **Aeolosoma** uses cilia around the mouth to sweep in food particles, and it buds off new individuals asexually. C, **Tubifex** lives head down in long tubes. D, **Dero** has ciliated anal gills.

- **Figure 17.22 Explanation:** Diversity of freshwater oligochaetes. **Stylaria** has a proboscis; **Tubifex** lives in tubes and is an indicator of pollution/low oxygen.

Class Hirudinida: Leeches

- Orders: **Hirudinea** (“true” leeches), **Acanthobdellida**, **Branchiobdellida**.
- Mostly freshwater, some marine or terrestrial.
- Fixed number of segments (usually 34).
- Lack setae (except Acanthobdellida).
- Possess anterior and posterior **suckers**.



Figure 23: The world's largest leech, **Haementeria ghilianii**, on the arm of Dr. Roy K. Sawyer, who found it in French Guiana, South America.

- **Figure 17.23 Explanation:** **Haementeria ghilianii**, the giant Amazonian leech.

Form and Function

- Dorsoventrally flattened.
- Segments marked by transverse grooves (**annuli**).
- No distinct coelomic compartments (septa lost). Coelom filled with connective tissue and **lacunae** (channels).
- Locomotion: Looping movements using suckers or swimming.



Figure 24: Structure of a leech, **Placobdella**. A, External appearance, dorsal view. B, Internal structure, ventral view.

- **Figure 17.24 Explanation:** Anatomy of a leech (**Placobdella**). Note the anterior and posterior suckers, the gut with ceca (for storage), and the lack of setae.

Nutrition

- Many are predaceous (carnivores); some are temporary or permanent parasites.
- Fluid feeders (blood, tissue fluids).
- **Hirudo medicinalis** (medicinal leech): True bloodsucker with chitinous jaws. Used medically for microsurgery (relieving venous congestion).



Figure 25: **Hirudo medicinalis** feeding on blood from human arm.

- **Figure 17.25 Explanation: Hirudo medicinalis** feeding. It uses cutting jaws to pierce skin and secretes anticoagulants.

Other Systems

- Respiration: Through skin (some fish leeches have gills).
- Excretion: Nephridia (10-17 pairs).
- Nervous System: Two “brains” (anterior and posterior fused ganglia) and ventral nerve cord. **Sensillae** (sense organs) in central annulus of each segment.
- Reproduction: Hermaphroditic; cross-fertilization; clitellum secretes cocoon.

Phylum Echiura

- “Spoon worms”. Marine, burrowing.
- Cylindrical body with a flattened, extensible **proboscis** anterior to the mouth (cannot be retracted).
- **No segmentation.**



Figure 26: A, **Echiurus**, an echiuran common on both Atlantic and Pacific coasts of North America. B, **Anelassorhynchus**, an echiuran of the tropical Pacific. The shape of their proboscis lends them the common name of “spoon worms.”

- **Figure 17.26 Explanation:** External morphology of Echiurans showing the characteristic non-retractable proboscis.
- Feeding: Deposit feeders. Proboscis extends to gather detritus via ciliated groove.
 - **Urechis:** Uses mucous net for filter feeding.



Figure 27: **Bonellia** (phylum Echiura) is a detritus feeder. Lying in its burrow, it explores the surface with its long proboscis, which picks up organic particles and carries them along a ciliated groove to the mouth.

- **Figure 17.27 Explanation:** **Bonellia** feeding with its extremely long proboscis.
- Anatomy: Muscular body wall, large coelom, long coiled gut, pair of **anal sacs** (excretion/osmoregulation).

- Reproduction: Separate sexes. **Bonellia** exhibits extreme sexual dimorphism (males are tiny and live on females). Sex determination is environmental (larvae settling on female become male).



Figure 28: Internal anatomy of an echiuran.

- **Figure 17.28 Explanation:** Internal anatomy of an echiuran, showing the coiled intestine and anal vesicles.

Phylum Sipuncula

- “Peanut worms”. Benthic marine.
- **No segmentation** or setae.
- Slender, retractile **introvert** (proboscis) with tentacles at the tip.



Figure 29: Sipunculans. **Themiste** (A) and **Phascolosoma** (B) are both burrowing genera of cosmopolitan distribution.

- **Figure 17.29 Explanation:** Sipunculans (**Themiste**, **Phascolosoma**) showing the trunk and the introvert.
- Feeding: Deposit or suspension feeders. Tentacles collect food.
- Anatomy: U-shaped digestive tract (anus near base of introvert). **Compensation sacs** connected to tentacles. Large fluid-filled coelom.
- Reproduction: Separate sexes; trochophore larva.



Figure 30: Internal structure of **Sipunculus**.

- **Figure 17.30 Explanation:** Internal structure of **Sipunculus**. Note the U-shaped gut and the retractor muscles for the introvert.

Evolutionary Significance of Metamerism

- **Hydrostatic Skeleton:** Coelom facilitates burrowing. Segmentation allows independent muscle contraction in separate segments, increasing efficiency.
- Redundancy: Safety factor if parts fail.
- Metamerism likely arose independently in Annelida, Arthropoda, and Chordata.

Phylogeny and Adaptive Diversification

- **Phylogeny:**
 - Annelids and Molluscs are closely related (Lophotrochozoa).
 - Siboglinids are derived polychaetes.
 - Echiurans and Sipunculans are closely related to Annelids; Echiurans may be derived polychaetes that lost segmentation.
- **Adaptive Diversification:**
 - Basic annelid body plan (coelom, segmentation) allows for diverse modifications.
 - Polychaetes: Parapodia adaptations (swimming, respiration, feeding).
 - Oligochaetes: Adaptations for burrowing and terrestrial life.
 - Leeches: Adaptations for predation and blood-sucking (suckers, jaws).